SUPPLEMENT TO "AN EXACT t-TEST"

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1. ADDITIONAL PROOFS AND TECHNICAL RESULTS

1.1. *Proof of Theorem 1*

The proof is given for the general case of $\mathbf{X}_1 \in \mathbb{R}^{p_1}$, $p_1 \leq p$. Furthermore, results herein immediately extend to the case of testing a general linear hypothesis $R\beta = 0$ for any given full rank matrix R. See Lei and Bickel (2021).

PROOF OF THEOREM 1: We omit the subscript and use $\mathbf{G} = \mathbf{G}_n$ and $\mathcal{Q} = \mathcal{Q}_{\mathbf{G}\mathbf{X}_2}$ for ease of notation. First, observe that, under the null and invoking Lemma A.1, the numerator rewrites as

$$\mathbf{X}_1^T \left(\mathcal{Q} g \left(\mathbf{Y} - \mathbf{X}_1 \beta_1^0\right)\right) = \mathbf{X}_1^T \left(g \mathcal{Q} \left(\mathbf{Y} - \mathbf{X}_1 \beta_1^0\right)\right) = \mathbf{X}_1^T \mathcal{Q} \left(g\boldsymbol{\varepsilon}\right), \tag{1}$$

which is a function of $(\mathbf{X}_1, \mathbf{X}_2, g\boldsymbol{\varepsilon})$ alone.

Second, let $\hat{\mathcal{Q}}$ be the orthogonal projection onto $\mathcal{C}^{\perp}\left(\mathbf{1}_n, \{g\mathbf{X}_1\}_{g\in\mathbf{G}}, \{g\mathbf{X}_2\}_{g\in\mathbf{G}}\right)$, the orthogonal complement of the span of $\mathbf{1}_n$, $\{g\mathbf{X}_1\}_{g\in\mathbf{G}}$ and $\{g\mathbf{X}_2\}_{g\in\mathbf{G}}$. Consider the permuted fitted residuals $\hat{\boldsymbol{\varepsilon}}(g) \equiv \hat{\boldsymbol{\varepsilon}}\left(\mathbf{X}_1, \mathbf{X}_2, g\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)\right)$, the residuals from the regression of $g\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)$ onto $\mathbf{1}_n$, $\{g\mathbf{X}_1\}_{g\in\mathbf{G}}$ and $\{g\mathbf{X}_2\}_{g\in\mathbf{G}}$. Observe that $\hat{\boldsymbol{\varepsilon}}(g)$ rewrites as

$$\hat{\mathcal{Q}} g \left(\mathbf{Y} - \mathbf{X}_1 \beta_1^0\right) = g\hat{\mathcal{Q}} \left(\mathbf{Y} - \mathbf{X}_1 \beta_1^0\right) = g\hat{\mathcal{Q}}\boldsymbol{\varepsilon} = \hat{\mathcal{Q}} g\boldsymbol{\varepsilon}, \tag{2}$$

where the first and third equality invoke Lemma A.1. In particular, $\hat{\mathcal{Q}}\boldsymbol{\varepsilon} = \hat{\boldsymbol{\varepsilon}}(\mathrm{id})$ and we deduce that $g\hat{\boldsymbol{\varepsilon}}(\mathrm{id}) \equiv (\hat{\varepsilon}_{g(1)}, \ldots, \hat{\varepsilon}_{g(n)})$ is equal to $\hat{\boldsymbol{\varepsilon}}(g)$ which is a function of $(\mathbf{X}_1, \mathbf{X}_2, g\boldsymbol{\varepsilon})$.

Standard arguments then apply to establish equation (6) from the main text. Specifically, consider the explicit parametrization of the test

$$\phi\left(\mathbf{W}, \{g\mathbf{W}'\}_{g\in\mathbf{G}}\right) = \begin{cases} 1, & \hat{t}\left(\mathbf{W}\right) > t^{(k)}\left(\mathbf{W}'\right) \\ a, & \hat{t}\left(\mathbf{W}\right) = t^{(k)}\left(\mathbf{W}'\right) \\ 0, & \hat{t}\left(\mathbf{W}\right) < t^{(k)}\left(\mathbf{W}'\right) \end{cases},$$

where $\hat{t}\left(\mathbf{W}\right) = t\left(\beta_1^0 : \mathbf{W}\right)$ and $t^{(k)}\left(\mathbf{W}'\right)$ is the kth order statistic of $\{t(\beta_1^0 : g\mathbf{W}')\}_{g\in\mathbf{G}}$. In particular, $\phi\left(\mathbf{W}\right) \equiv \phi\left(\mathbf{W}, \{g\mathbf{W}\}_{g\in\mathbf{G}}\right)$.

By construction,

$$\sum_{g\in\mathbf{G}} \phi\left(g\mathbf{W}\right) \equiv \sum_{g\in\mathbf{G}} \phi\left(g\mathbf{W}, \{g'g\mathbf{W}\}_{g'\in\mathbf{G}}\right) = \sum_{g\in\mathbf{G}} \phi\left(g\mathbf{W}, \{g\mathbf{W}\}_{g\in\mathbf{G}}\right) = M\alpha,$$

where the second equality holds because $\mathbf{G}$ is a group. Therefore,

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$$M\alpha = E\left[\sum_{g\in\mathbf{G}} \phi\left(g\mathbf{W}\right)\right] = \sum_{g\in\mathbf{G}} E\left[\phi\left(g\mathbf{W}\right)\right].$$

The Strong Null then implies that

$$M\alpha = \sum_{g\in\mathbf{G}} E\left[\phi\left(\mathbf{W}\right)\right] = ME\left[\phi\left(\mathbf{W}\right)\right].$$

Q.E.D.

1.2. *Sufficient conditions for Assumption S*

Emulating the definition of the stochastic upper bound, which states that $X_n = O_P\left(a_n\right)$ if, for any $\epsilon > 0$, there exists M and N such that $P\left(\left|\frac{X_n}{a_n}\right| > M\right) < \epsilon$, $\forall\ n \geq N$, we may refer to the stochastic lower bound, which may be expressed as $X_n = \Omega_P\left(b_n\right)$ if, for any $\epsilon > 0$, there exists c and N such that $P\left(\frac{X_n}{b_n} > c\right) \geq 1 - \epsilon$, $\forall\ n \geq N$.

Importantly, the two are related. In particular, if $X_n = \Omega_P\left(a_n\right)$, then $X_n^{-1} = O_P\left(1/a_n\right)$.[1]

LEMMA SA.1: *If $n \geq \overrightarrow{p}$ and $\mathbf{X}_2 \in \mathbb{R}^{n\times(p-1)}$, the design matrix of nuisance regressors, has independent entries $X_{i,2} \in \mathbb{R}^{p-1}$ whose distribution is uniformly continuous with respect to Lebesgue measure, then $\mathbf{G}\mathbf{X}_2$ achieves the upper bound on its rank, i.e., $\operatorname{rank}\left(\mathbf{G}\mathbf{X}_2\right) = (p-1)\cdot\left(n_{\text{blocks}}\cdot\left(n_{\text{blocks}}-2\right)+2\right)$ with probability one.*

This is immediate from the proof of Theorem A.1 and is stated without proof.

LEMMA SA.2: *Suppose $\mathbf{X}_2 \in \mathbb{R}^{n\times(p-1)}$, the design matrix of nuisance regressors, has independent and identically distributed rows $X_{i,2} \in \mathbb{R}^{p-1}$ whose distribution is absolutely continuous with respect to Lebesgue measure and let $n \geq \overrightarrow{p}$. Then,*

$$\lambda_{\max}\left(\left(\frac{\overrightarrow{\mathbf{X}}^T\overrightarrow{\mathbf{X}}}{n}\right)^{-1}\right) = O_P\left(\overrightarrow{p}\right).$$

PROOF: Write $\overrightarrow{\mathbf{X}}^T \in \mathbb{R}^{\overrightarrow{p}\times n}$, where $\overrightarrow{p} = (p-1)\cdot\left(n_{\text{blocks}}\cdot\left(n_{\text{blocks}}-2\right)+2\right)$, as

$$\begin{pmatrix} A_1 \cdots A_b\ B_1 \cdots B_b \cdots \cdots Z_b \\ B_1 \cdots B_b\ A_1 \cdots A_b \cdots \cdots Y_b \\ \vdots \\ Z_1 \cdots Z_b\ Y_1 \cdots Y_b \cdots \cdots A_b \end{pmatrix},$$

[1] If $X_n = \Omega_P\left(b_n\right)$, then for any $\epsilon > 0$, there exists c and N such that $P\left(\left|\frac{X_n}{b_n}\right| > c\right) \geq P\left(\frac{X_n}{b_n} > c\right) \geq 1 - \epsilon$,which rearranges as $P\left(\left|\frac{X_n}{b_n}\right| \leq c\right) < \epsilon$. This means that $P\left(\left|\frac{X_n^{-1}}{1/b_n}\right| > c^{-1}\right) < \epsilon$.

where each $A_j, B_j, \ldots$, $j = 1, \ldots, b$ is a $(p-1) \times 1$ matrix. Specifically, we may write those out in terms of entries

$$A_j = (A_{j,1}, \ldots, A_{j,p-1})^T = (X_{2,j}, \ldots, X_{p-1,j})^T,$$
$$B_j = (B_{j,1}, \ldots, B_{j,p-1})^T = (X_{2,b+j}, \ldots, X_{p-1,b+j})^T,$$

and so on.

Define the $\overrightarrow{p} \times (\underline{b} \cdot n_{\text{blocks}})$ submatrices

$$\overline{\mathbf{X}}_1 = \begin{pmatrix} A_1 \cdots A_{\underline{b}} \; B_1 \cdots B_{\underline{b}} \cdots \cdots Z_{\underline{b}} \\ B_1 \cdots B_{\underline{b}} \; A_1 \cdots A_{\underline{b}} \cdots \cdots Y_{\underline{b}} \\ \vdots \\ Z_1 \cdots Z_{\underline{b}} \; Y_1 \cdots Y_{\underline{b}} \cdots \cdots A_{\underline{b}} \end{pmatrix}, \overline{\mathbf{X}}_2 = \begin{pmatrix} A_{\underline{b}+1} \cdots A_{2\underline{b}} \; B_{\underline{b}+1} \cdots B_{2\underline{b}} \cdots \cdots Z_{2\underline{b}} \\ B_{\underline{b}+1} \cdots B_{2\underline{b}} \; A_{\underline{b}+1} \cdots A_{2\underline{b}} \cdots \cdots Y_{2\underline{b}} \\ \vdots \\ Z_{\underline{b}+1} \cdots Z_{2\underline{b}} \; Y_{\underline{b}+1} \cdots Y_{2\underline{b}} \cdots \cdots A_{2\underline{b}} \end{pmatrix},$$

under the null, where $\underline{b} = \lceil \overrightarrow{p} / n_{\text{blocks}} \rceil$.[2] Finally, if $b/\underline{b}$ is not an integer, let

$$\overline{\mathbf{X}}_{\lfloor b/\underline{b} \rfloor + 1} = \begin{pmatrix} A_{\lfloor b/\underline{b} \rfloor \underline{b}+1} \cdots A_b \; B_{\lfloor b/\underline{b} \rfloor \underline{b}+1} \cdots B_b \cdots \cdots Z_b \\ B_{\lfloor b/\underline{b} \rfloor \underline{b}+1} \cdots B_b \; A_{\lfloor b/\underline{b} \rfloor \underline{b}+1} \cdots A_b \cdots \cdots Y_b \\ \vdots \\ Z_{\lfloor b/\underline{b} \rfloor \underline{b}+1} \cdots Z_b \; Y_{\lfloor b/\underline{b} \rfloor \underline{b}+1} \cdots Y_b \cdots \cdots A_b \end{pmatrix}.$$

Specifically, up to reordering of the rows of $\overrightarrow{\mathbf{X}}$, we have $\overrightarrow{\mathbf{X}}^T = \left(\overline{\mathbf{X}}_1, \overline{\mathbf{X}}_2, \ldots, \overline{\mathbf{X}}_{\lfloor n/\underline{b} \rfloor + 1}\right)$. We can represent the matrix inner product $\overrightarrow{\mathbf{X}}^T \overrightarrow{\mathbf{X}}$ as a sum of outer products of these submatrices,

$$\overrightarrow{\mathbf{X}}^T \overrightarrow{\mathbf{X}} = \sum_{i=1}^{\lfloor n/\underline{b} \rfloor + 1} \overline{\mathbf{X}}_i \overline{\mathbf{X}}_i^T.$$

Applying Weyl's inequality (Bhatia, 2013, Theorem III.2.1), we find that

$$\lambda_{\min}\left(\overrightarrow{\mathbf{X}}^T \overrightarrow{\mathbf{X}}\right) \geq \sum_{i=1}^{\lfloor b/\underline{b} \rfloor + 1} \lambda_{\min}\left(\overline{\mathbf{X}}_i \overline{\mathbf{X}}_i^T\right) \geq \sum_{i=1}^{\lfloor b/\underline{b} \rfloor} \lambda_{\min}\left(\overline{\mathbf{X}}_i \overline{\mathbf{X}}_i^T\right).$$

Remark that, for $i = 1, \ldots, \lfloor b/\underline{b} \rfloor$, $\overline{\mathbf{X}}_i$ is a design matrix of nuisance corresponding to the case with $n_{\text{blocks}} \cdot \underline{b}$ observations. Because $n_{\text{blocks}} \cdot \underline{b} \geq \overrightarrow{p}$ and the observations are assumed to be continuous random variables, by Lemma SA.1 we know that $\overline{\mathbf{X}}_i$ is full rank and thus $\lambda_{\min}\left(\overline{\mathbf{X}}_i \overline{\mathbf{X}}_i^T\right) > 0$ with probability 1. Further remark that each $\overline{\mathbf{X}}_i$ is made of distinct rows of the original data matrix $\mathbf{X}_2$, meaning they are independent from each other and identically distributed, and hence so are the random eigenvalues $\lambda_{\min}\left(\overline{\mathbf{X}}_i \overline{\mathbf{X}}_i^T\right)$, $i = 1, \ldots, \lfloor n/\underline{b} \rfloor$.

Consequently,

$$\sum_{i=1}^{\lfloor b/\underline{b} \rfloor} \lambda_{\min}\left(\overline{\mathbf{X}}_i \overline{\mathbf{X}}_i^T\right) = \Omega_P\left(\lfloor b/\underline{b} \rfloor\right),$$

[2]For intuition, the choice of $\underline{b}$ is meant to insure that each $\overline{\mathbf{X}}_i$ corresponds to the design matrix of a regression problem with at least $\overrightarrow{p}$ observations, thus ensuring it is a full rank design matrix with probability one by Lemma SA.1 and Theorem A.1, while having as few "observations" as possible so to extract as many submatrices as possible.

TABLE I

PROBABILITY OF REJECTION OF THE NULL AT NOMINAL LEVEL $\alpha = 0.1$

Block size	1	10	20	50	1	10	20	50	1	10	20	50
$n = 100$	0.1	0.1	0.1	0	0.43	0.42	0.38	0	1	1	1	0
$n = 200$	0.1	0.1	0.1	0.08	0.61	0.60	0.58	0.45	1	1	1	1
$n = 400$	0.1	0.1	0.1	0.1	0.79	0.79	0.78	0.74	1	1	1	1
$n = 800$	0.1	0.1	0.1	0.1	0.95	0.95	0.94	0.94	1	1	1	1

Note: Columns are block size. Errors of the posited nulls are $\beta_1 - \beta_1^0 = 0$ (left), $\beta_1 - \beta_1^0 = 0.1$ (center), $\beta_1 - \beta_1^0 = 1$ (right). We use the "non-randomized" test here. Critical level is $\alpha = 0.1$.

using the law of large numbers. Thus

$$\lambda_{\max}\left(\left(\frac{\overrightarrow{\mathbf{X}}^T\overrightarrow{\mathbf{X}}}{n}\right)^{-1}\right) = O_P\left(\overrightarrow{p}\right),$$

where we used that if a random term is $O_P\left(n/\left\lfloor b/\underline{b}\right\rfloor\right)$, then it is $O_P\left(\overrightarrow{p}\right)$.

Q.E.D.

2. IMPACT ON POWER OF USING BLOCK PERMUTATIONS IN THE UNIVARIATE REGRESSION MODEL

The univariate case is

$$\mathbf{Y} = \beta_0 \mathbf{1}_n + \mathbf{X}\beta_1 + \boldsymbol{\varepsilon},$$

where $\mathbf{X} \in \mathbb{R}^n$ is the univariate regressor.

The test statistic is

$$t_g\left(\beta_1^0\right) = \frac{1}{\sqrt{n}}\left(g\mathbf{X}\right)^T\left(\mathbf{Y} - \mathbf{X}\beta_1^0\right).$$

The key is that this framework allows us to investigate the loss of power, or lack thereof, from using a smaller group in a way that is "isolated". Indeed, in a general set-up, smaller groups also increase power through a different channel, i.e., they conserve more explanatory variation in $\mathcal{Q}\mathbf{X}_1$. But in the univariate setting, there are no covariates to orthogonalize. Consequently, potential loss of power from using fewer blocks is not counterbalanced by a gain in power due to weaker orthogonalization of nuisance covariates, and can be studied on its own.

We can see in Table I simulation evidence that the loss of power from using block permutations is apparently of small magnitude.

3. SHORT SURVEY OF EXACT TESTS FOR SUBVECTORS

Different approaches have been suggested to produce exact tests for subvectors in the linear regression model. At the time of this writing, the only other asymptotically robust approach allowing for dependence between covariates is that of D'Haultfœuille and Tuvaandorj (2024), which requires that covariates which are not being tested be discrete.

The original proposition of Lei and Bickel (2021) in fact offers a general constrained optimization framework which, if slightly generalized, can be seen as producing –a generalization of– the unstudentized version of our proposed method as a special case. See Subsection 3.1.

The alternative approaches of Wen et al. (2022) and Guan (2023) may be seen through the framework of this paper. See Subsection 3.2.

3.1. *Unifying framework of Lei and Bickel (2021)*

Lei and Bickel (2021) produce an exact test by imposing invariance as feasibility conditions and then optimizing for the feasible test statistic with the greatest power according to natural criteria of sensitivity to departure from the null.

Their suggested procedure is specific to cyclical permutations. However, their framework – constrain for feasibility then optimize for power– accommodates much more general invariance criteria.

In fact, we can cast the unstudentized version of our proposed test statistic as a nontrivial feasible solution in the generalized framework of Lei and Bickel (2021) when the invoked invariance is with respect to block permutations, as opposed to cyclical permutations.

We present their framework, pointing out its generality. We explain how, when applied to the block permutation invariance criteria, the exact t-test is a feasible solution.

We can state the general framework almost exactly in the notation of Lei and Bickel (2021). For ease of exposition, we consider the null $H_0 : \beta_1 = 0$.

To better align with the notation of Lei and Bickel (2021), let Π_j be the matrix representations of the linear actions g_j, $j = 0, 1, \ldots, m$.

In this general framework, we construct a set of test statistics of the form

$$S_j = \eta_j^T \mathbf{Y} = (\Pi_j \eta_0)^T \mathbf{Y},$$

for some $\eta_0 \in \mathbb{R}^n$, where $\eta_j = \Pi_j \eta_0$, $\Pi_0 = I_n$, and $\{\Pi_0, \Pi_1, \ldots, \Pi_m\} = \mathbf{G}_n$ forms an algebraic group.

Two conditions are imposed to guarantee the invariance of the test statistic.

Condition 1 is that the "deterministic" part of the regression equation be constant –though not necessarily zero– across applications of the permutations:

$$\mathbf{X}_2^T \eta_j = \gamma_2, \ j = 0, 1, \ldots, m, \tag{3}$$

for some $\gamma_2 \in \mathbb{R}^{p-1}$.

Condition 2 is that the group actions leave the distribution of the error term ε invariant. Hence, we require that

$$\eta_j = \Pi_j \eta_0, \ j = 1, \ldots, m, \tag{4}$$

for some $\eta_0 \in \mathbb{R}^n$.

Then conditions 1 and 2 have representation

$$\begin{pmatrix} -I_{p-1} & \mathbf{X}_2^T \\ -I_{p-1} & \mathbf{X}_2^T \Pi_1 \\ \vdots & \vdots \\ -I_{p-1} & \mathbf{X}_2^T \Pi_m \end{pmatrix} \begin{pmatrix} \gamma_2 \\ \eta_0 \end{pmatrix} = 0.$$

A third condition requires that the procedure have power to detect departures from the null. Specifically, we impose

$$\mathbf{X}_1^T \eta_j \leq \gamma_1, \ j = 1, 2, \ldots, m, \text{ and } \mathbf{X}_1^T \eta_0 = \gamma_1 + \delta, \tag{5}$$

for some scalars γ_1 and δ.[3] If δ is large, then the procedure is "powerful" by design in this specific, implied sense.

[3]The exact t-test framework accommodates (5) because we could set $\eta_0 = g^* \overline{\mathbf{X}}_1$ for $g^* = \max_g \mathbf{X}_1^T \left(g \overline{\mathbf{X}}_1\right)$.

Anticipating a constrained optimization problem for which they want to produce a closed form solution, Lei and Bickel (2021) impose a stronger requirement, that

$$\mathbf{X}_1^T \eta_j = \gamma_1,\ j = 1, 2, \ldots, m, \text{ and } \mathbf{X}_1^T \eta_0 = \gamma_1 + \delta. \tag{6}$$

Combining conditions (3), (4) and (6), and optimizing for the largest power, as measured by δ, we obtain the constrained program

$$\max_{\delta \in \mathbb{R}, \gamma \in \mathbb{R}^p, \eta_0 \in \mathbb{R}^n} \delta$$

subject to

$$\begin{pmatrix} e_{1,p} & -I_p & \mathbf{X}^T \\ \mathbf{0}_p & -I_p & \mathbf{X}^T \Pi_1 \\ \vdots & \vdots & \vdots \\ \mathbf{0}_p & -I_p & \mathbf{X}^T \Pi_m \end{pmatrix} \begin{pmatrix} \delta \\ \gamma \\ \eta_0 \end{pmatrix} = 0,$$

$$\|\eta_0\|_2 = 1,$$

where $\gamma^T = (\gamma_1, \gamma_2^T)$ and $e_{1,p}^T = (1, 0, \ldots, 0) \in \mathbb{R}^p$.

Theorem 2 of Lei and Bickel (2021) applies to this problem with general linear actions $I, \Pi_1, \ldots, \Pi_m$ and delivers the following closed form. Suppose $n \geq (p-1) \cdot m$ and let

$$B(X) = \left((I - \Pi_m)^T \mathbf{X}_2, (\Pi_1 - \Pi_m)^T \mathbf{X}_2, \ldots, (\Pi_{m-1} - \Pi_m)^T \mathbf{X}_2 \right) \in \mathbb{R}^{n \times m(p-1)}.$$

This can be subdivided as

$$B(X) = \left(B(X)_1, B(X)_{[-1]} \right),$$

where $B(X)_1$ is the first column of $B(X)$. Further define

$$\tilde{\eta} = (I - H_{[-1]}) B(X)_1,$$

where

$$H_{[-1]} = B(X)_{[-1]} \left(B(X)_{[-1]}^T B(X)_{[-1]} \right)^- B(X)_{[-1]}^T$$

and the "$-$" indicates the Moore-Penrose inverse. The solution of the above program is

$$\eta_0^* = \frac{\tilde{\eta}}{\|\tilde{\eta}\|_2}.$$

This completes the construction of the first exact test for subvectors in linear regression, as detailed in Lei and Bickel (2021).

If we use the less constraining power condition (5) instead of (6), we obtain a weakly greater optimal objective function evaluation but have to forgo the closed form solution afforded by Theorem 2 of Lei and Bickel (2021). The resulting constrained optimization program is

$$\max_{\delta \in \mathbb{R}, \gamma \in \mathbb{R}^p, \eta_0 \in \mathbb{R}^n} \delta$$

subject to

$$\begin{pmatrix} -e_{1,p} & -I_p & \mathbf{X}^T \\ \mathbf{0}_p & -I_p & \mathbf{X}^T\Pi_1 \\ \vdots & \vdots & \vdots \\ \mathbf{0}_p & -I_p & \mathbf{X}^T\Pi_m \end{pmatrix} \begin{pmatrix} \delta \\ \gamma_1 \\ \gamma_2 \\ \eta_0 \end{pmatrix} \begin{pmatrix} =_p \\ \lessdot_p \\ \vdots \\ \lessdot_p \end{pmatrix} 0,$$

$$\|\eta_0\|_2 = 1,$$

where $=_p$ stands for the p-tuple $(=,\ldots,=)$ and $\lessdot_p$ stands for the p-tuple $(\leq,=,\cdots=)$.

This formulation has our proposed statistic as a feasible solution. In our simulation studies, when implementing this optimized approach with non-overlapping block permutations, we did not witness systematic gains in power.

3.2. *Alternative Approaches*

The construction of Wen et al. (2022) can be understood as follows. In order to deliver the desired invariance, the authors suggest taking a worst case over all permutations. Their proposed test statistics are

$$T_{\mathrm{id}}^{\mathrm{RPT}}\left(\beta_1^0\right) = \min_{g^*}\left\{\mathbf{X}_1^T\mathcal{Q}_{g^*}\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right)\right\} \overset{H_0}{=} \min_{g^*}\left\{\mathbf{X}_1^T\mathcal{Q}_{g^*}\boldsymbol{\varepsilon}\right\},$$

and the oracle statistic

$$T_g^{\mathrm{RPT}}\left(\beta_1^0\right) = T^{\mathrm{RPT}}\left(\beta_1^0;\mathbf{X}_1,\mathbf{X}_2,g\boldsymbol{\varepsilon}\right) = \min_{g^*}\left\{\mathbf{X}_1^T\mathcal{Q}_{g^*}g\boldsymbol{\varepsilon}\right\},\ g\neq \mathrm{id},$$

where $\mathcal{Q}_g$ is an orthogonal projection onto (a subspace of) $\mathcal{C}\left(\mathcal{Q}_{\mathbf{X}_2}\right)\cap\mathcal{C}\left(\mathcal{Q}_{g\mathbf{X}_2}g\right)$. This statistic is directly defined as a function of $(\mathbf{X}_1,\mathbf{X}_2,g\boldsymbol{\varepsilon})$ and exact validity using this statistic thus obtains under the Strong Null but it is not computable using observed data. It may however be bounded using a statistic which is computable.

Consider the test statistic

$$T_g^{\overline{\mathrm{RPT}}}\left(\beta_1^0\right) = T_g^{\overline{\mathrm{RPT}}}\left(\beta_1^0;\mathbf{X}_1,\mathbf{X}_2,g\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right)\right) = \mathbf{X}_1^T\mathcal{Q}_g g\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right) \overset{H_0}{=} \mathbf{X}_1^T\mathcal{Q}_g g\boldsymbol{\varepsilon},$$

where, obviously, $T_g^{\mathrm{RPT}}\left(\beta_1^0\right)\leq T_g^{\overline{\mathrm{RPT}}}\left(\beta_1^0\right)$. Remarking that

$$\hat{p} := \frac{1}{M}\sum_{g\in\mathbf{G}_n}\mathbf{1}\left\{T_{\mathrm{id}}^{\mathrm{RPT}}\left(\beta_1^0\right)\leq T_g^{\mathrm{RPT}}\left(\beta_1^0\right)\right\} \leq \frac{1}{M}\sum_{g\in\mathbf{G}_n}\mathbf{1}\left\{T_{\mathrm{id}}^{\mathrm{RPT}}\left(\beta_1^0\right)\leq T_g^{\overline{\mathrm{RPT}}}\left(\beta_1^0\right)\right\} =: \tilde{p},$$

we can use $\tilde{p}$ as a conservative estimate of the p-value $\hat{p}$. Specifically, a test with guarantee of probability of type 1 error α that uses a threshold of α for $\tilde{p}$ as its accept-reject decision will be conservative but may be computed from data.

Guan (2023) proposes a different procedure based on approximating p-values. She suggests using as an estimated p-value the statistic

$$\hat{P}\left(\beta_1^0\right) = \frac{1}{M}\sum_{g^*\in\mathbf{G}_n}\mathbf{1}\Big\{\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right)^T\mathcal{Q}_{X_1,X_2,g^*X_2}\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right) \geq \left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right)^T\mathcal{Q}_{g^*X_1,g^*X_2,X_2}\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right)\Big\}$$

$$\overset{H_0}{=} \frac{1}{M} \sum_{g^* \in \mathbf{G}_n} \mathbf{1}\left\{\varepsilon^T \mathcal{Q}_{X_1, X_2, g^* X_2} \varepsilon \geq \varepsilon^T \mathcal{Q}_{g^* X_1, g^* X_2, X_2} \varepsilon\right\}.$$

Guan (2023) constructs an exactly valid test as follows. Define the oracle statistic

$$\hat{P}_g = \frac{1}{M} \sum_{g^* \in \mathbf{G}_n} \mathbf{1}\left\{\varepsilon^T \mathcal{Q}_{g X_1, g X_2, g^* X_2} \varepsilon \geq \varepsilon^T \mathcal{Q}_{g^* X_1, g^* X_2, g X_2} \varepsilon\right\},$$

and remark that $\hat{P}\left(\beta_1^0\right) = \hat{P}_{\mathrm{id}}\left(\beta_1^0\right)$.

Consider the $M \times M$ matrix

$$W \equiv \{W_{g,g^*}\}_{g,g^*} := \left\{\mathbf{1}\left\{\varepsilon^T \mathcal{Q}_{g X_1, g X_2, g^* X_2} \varepsilon \geq \varepsilon^T \mathcal{Q}_{g^* X_1, g^* X_2, g X_2} \varepsilon\right\}\right\}_{g,g^*},$$

where the indicator takes value $1/2$ in case of a tie. Note that $\hat{P}_g\left(\beta_1^0\right) = 1/M \cdot \sum_{g^*} W_{g,g^*}$. Observing $W_{g,g^*} \geq 0$, $W_{g,g} = 1$, and $W_{g,g^*} + W_{g^*,g} = 1$, Guan (2023) calls upon a counting argument from multisplit conformal inference which guarantees that

$$\frac{1}{M} \# \left\{g : \frac{1}{M} \sum_{g^*} W_{g,g^*} \leq \alpha\right\} < 2\alpha. \tag{7}$$

Conditioning on the design matrix and the errors but not the order of the errors, i.e., $\varepsilon \sim U\{g\varepsilon\}_{g \in \mathbf{G}_n}$, we can see that

$$\hat{P}_g\left(\beta_1^0\right) \sim U\left\{\hat{P}_{g^{(1)}}\left(\beta_1^0\right), \ldots, \hat{P}_{g^{(M)}}\left(\beta_1^0\right)\right\}, \text{ for any given } g \in \mathbf{G}_n,$$

under the null, and therefore

$$P\left(\hat{P}_{\mathrm{id}}\left(\beta_1^0\right) \leq \alpha/2\right) = \frac{1}{M} \# \left\{g : \hat{P}_g\left(\beta_1^0\right) \leq \alpha/2\right\} \leq \alpha,$$

for any $\alpha \in [0, 1]$.

Consequently, $\hat{P}_{\mathrm{id}}\left(\beta_1^0\right)$ is an anti-conservative p-value estimate but if, say, we threshold the accept-reject decision at 0.05, we are guaranteed a probability of type one error of at most 0.1.

4. ASSESSING POWER VIA SIMULATIONS AND APPLICATIONS

4.1. *Simulations*

The objective here is to qualitatively substantiate the claim that the proposed test has reasonable power even though the requirements for validity have been weakened. In order to do that with simple yet compelling illustrations, we focus on *ex ante* expected data generating processes to impress upon the reader that the simulation designs were not cherry picked. Specifically, we consider standard normal errors for the "Gaussian" design, Gamma variables with shape and rate parameter equal to 0.01 and 1 and rescaled to have variance 1 for the "skewed" design, the "exponential" design corresponds to error distributed according to an exponential distribution with parameter 1, and normal errors with variance $\sigma_\epsilon^2(X) \propto X_1^2$ for the "heteroskedastic" design.

Unless otherwise specified, the covariates (X_{i1}, X_{i2}) are drawn from a bivariate normal with marginal variances 1 and covariance 0.15.

We can see from Figure 1 that, as expected, the classical t-test apparently dominates the proposed test in small samples with Gaussian data. However, the performance of the proposed

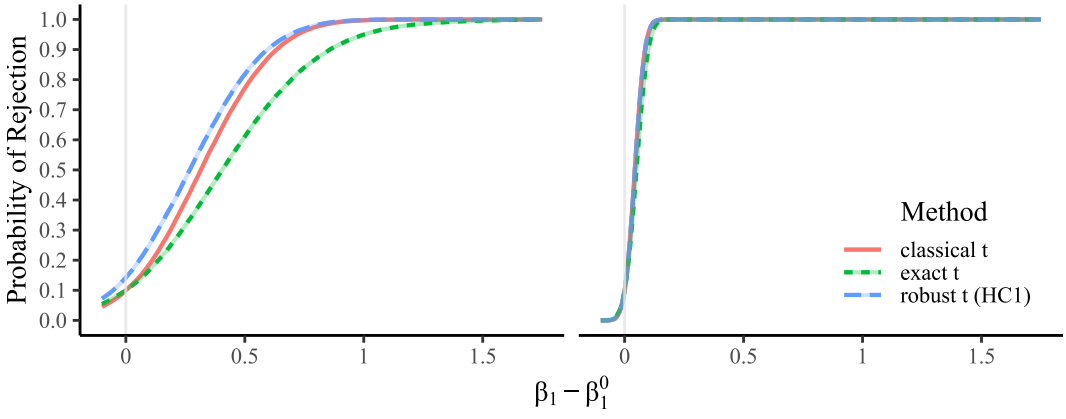


FIGURE 1.—Comparison of classical t-tests with proposed exact t-test when errors are normally distributed and critical level $\alpha = 0.1$. Left- and right-hand side plots are for $n = 25$ and 1000, and $n_{\text{blocks}} = 5$ and 7, respectively.

TABLE II

RELATIVE POWER OF PROPOSED TEST ALONG A PITMAN SEQUENCE.

n (n_{blocks})	25 (5)	50 (5)	100 (5)	1000 (7)
rel. to t-test, Gaussian errors	0.88	0.97	0.98	0.99
rel. to robust t-test, heteroskedastic errors	0.91	0.92	0.93	0.96

Note: The rejection probability of the proposed exact test divided by the rejection probability of the standard t-test. The tested null is $\beta_1^0 = \beta_1 + \sqrt{15/n}$. Errors are standard normal. Note that the power of the heteroskedasticity robust procedures (second row) may not be meaningful for $n \leq 100$ because these procedures are only asymptotically valid. Critical level is $\alpha = 0.1$.

test appears to be competitive asymptotically. This is illustred in the right-hand side panel of Figure 1 and quantified in Table II, which documents the seeming convergence of the power of the classical t-test and proposed test with Gaussian errors. The same convergence is observed for the heteroskedasticity robust t-test and the proposed test.

TABLE III

ASSESSING POWER AND VALIDITY

$\beta_1^0 - \beta_1$	Error distribution	unstudentized exact t-test	CPT	RPT
0	Gaussian	0.10	0.10	0.04
	Exponential	0.10	0.10	0.04
	Skewed	0.10	0.10	0.04
	Heteroskedastic	0.12	0.13	0.14
1	Gaussian	0.96	0.40	0.91
	Exponential	0.77	0.97	0.90
	Skewed	0.84	0.89	0.95

Note: $n = 25$ and $n_{\text{blocks}} = 5$. The design matrix has independent standard normal entries with pairwise correlation 0.15. Critical level is $\alpha = 0.1$.

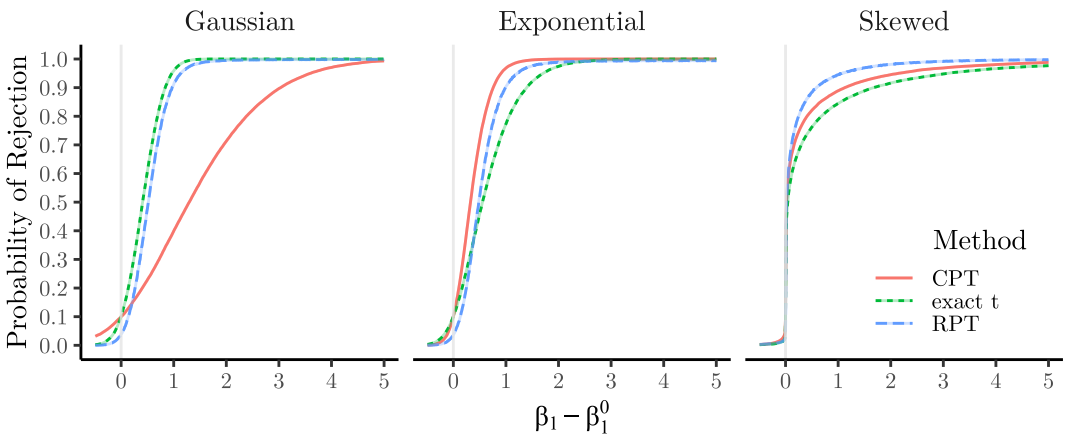


FIGURE 2.—Comparison of exact t-test with alternative methods in small sample. The bold black line corresponds to the proposed test, the blue dashed line corresponds to RPT, and the green dot dashed line corresponds to CPT. The left-hand side plot corresponds to Gaussian errors, the middle plot corresponds to exponential errors, and the right-hand side corresponds to the "skewed" design. Throughout, $n = 25$ and $n_{\text{blocks}} = 5$. Critical level is $\alpha = 0.1$.

TABLE IV

ASSESSING ASYMPTOTIC VALIDITY

$\beta_1^0 - \beta_1$	Error distribution	unstudentized exact t-test	CPT	RPT
0	Heteroskedastic	0.19	0.20	0.21

Note: $n = 2400$ and $n_{\text{blocks}} = 8$. The design matrix has independent standard normal entries with pairwise correlation 0.15. Critical level is $\alpha = 0.1$.

In the first three rows of Table III, we see that all exact tests are indeed valid in small samples with exchangeable errors.

In Figure 2 and in the last three rows of Table III, we see that neither the proposed procedure, CPT procedure of Lei and Bickel (2021), nor the residual permutation test (RPT) of Wen et al. (2022) dominates or is dominated by the others.

In Table III, we can examine the performance of candidate exact tests when the randomization hypothesis is violated. All three tests over-reject under heteroskedasticity.

As we can see from Table IV, under heteroskedasticity the size distortions do not vanish asymptotically for the unstudentized exact t-test, CPT, or RPT.

4.2. *Replication of an IV application*

Our method naturally extends to regression analysis with instrumental variables. Wooldridge (2020) reanalyzes Mroz (1987). He analyzes how $Y = \log(\text{wage})$ is affected by $X_1 = \text{education}$, instrumenting with $(Z_1, Z_2) = (\text{mother education}, \text{father education})$, controlling for $X_2 = \text{experience}$ and its square. In this case the Strong Null is

$$(\mathbf{Z}, \mathbf{X}_2, \varepsilon) \sim (\mathbf{Z}, \mathbf{X}_2, g\varepsilon)\,, \ \forall\, g \in \mathbf{G}_n.$$

Because there are two instruments, the natural test statistic is

$$t_g\left(\beta_1^0\right) = T_g\left(\beta_1^0\right)^T \hat{\Sigma}_g^{-1} T_g\left(\beta_1^0\right),$$

TABLE V

REANALYZING EXAMPLE 15.5 OF WOOLDRIDGE (2020).

$\hat{\beta}_{\text{educ}}$	randomized	p-val	classical	p-val	HC1	p-val
0.061	(0.010, 0.116)	0.075	(0.010, 0.113)	0.051	(0.006, 0.116)	0.066

Note: $n = 428$ and $n_{\text{blocks}} = 5$. Critical level for confidence intervals is $\alpha = 0.1$.

where $T_g\left(\beta_1^0\right) = \mathbf{Z}^T \mathcal{Q}\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)_g$ and $\hat{\Sigma}_g = \frac{1}{n}\sum_{i=1}^n \overline{\mathbf{Z}}_i \overline{\mathbf{Z}}_i^T \hat{\varepsilon}^2_{g(i)}$.

Remark the distinction with the instrumental variables analysis of Imbens and Rosenbaum (2005). They produce a Fisher test using the instrument assignment as a source of randomization and thus provide inference that is valid for the fixed, observed sample. In contrast, we rely on invariance under the null to produce inference that is valid unconditionally on the observed sample.

The confidence interval produced by the proposed method is presented in Table V. Its length is comparable to that of the confidence intervals produced by the standard inference methods for two-stage least-squares. Further note that the method is robust to weak identification of the tested coefficient. Of course, the method is compatible with the projection method if robustness to weak identification of nuisance coefficients is required.

4.3. *Replications of Wooldridge (2020) examples*

We replicate regression tables from the textbook Wooldridge (2020).

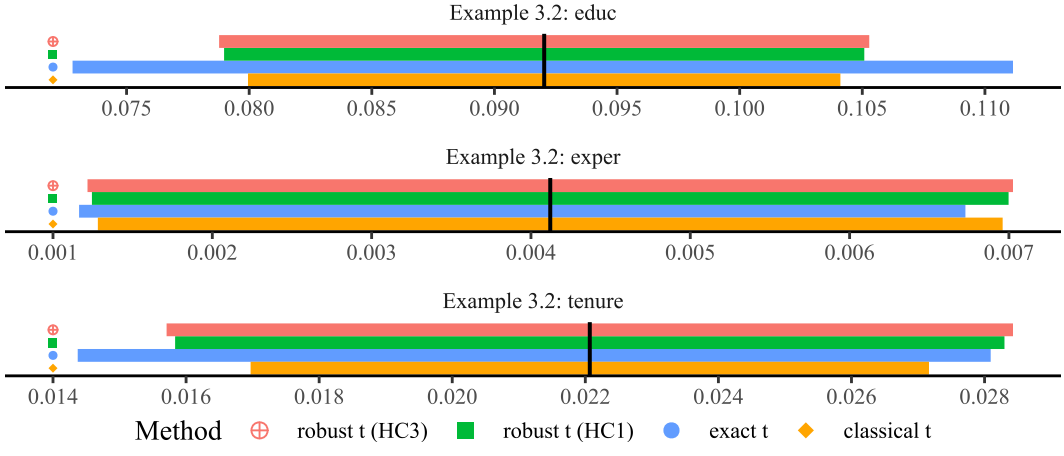


FIGURE 3.—Wooldridge (2020) example using "WAGE1" dataset. $n = 526$, $n_{blocks} = 5$ and $\alpha = 0.1$. The black vertical lines are the estimated coefficients.

5. COMPARISON OF ASYMPTOTIC REGIMES

A permutational central limit theorem (PCLT) establishes, informally speaking, the shape at which the histogram of the test statistics evaluated on permutations of the data stabilizes as n

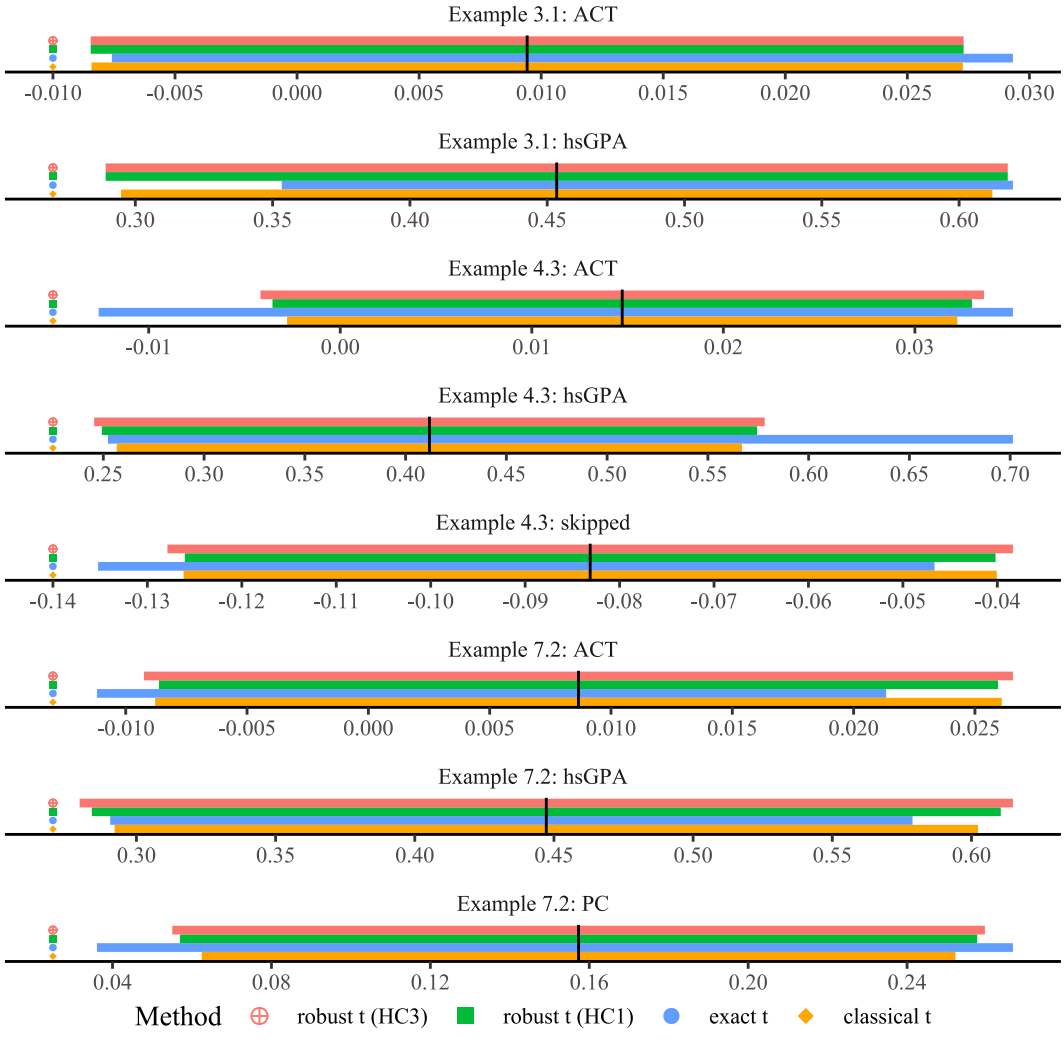


FIGURE 4.—Wooldridge (2020) examples using "GPA1" dataset. $n = 141$, $n_{blocks} = 5$ and $\alpha = 0.1$. The black vertical lines are the estimated coefficients.

gets large. Formally, it gives the limiting value of the randomization distribution, either almost surely or in probability.

One approach to establishing a PCLT is to show that, for any given non-degenerate data set, the histogram will converge to the same distribution. As such non-degenerate datasets occur with probability one in our applications of interest, almost sure convergence obtains.

Another approach, detailed in the Appendix of the main text, is to verify Hoeffding's criteria. This guarantees that the empirical distribution converges in probability to the limiting marginal distribution of a randomly permuted test statistic. The condition also has a nice "sharpness" to it; it is in fact necessary for convergence in probability of the empirical distribution (Chung and Romano, 2013).

In practice, while the almost sure convergence may feel more satisfying, it is in fact somewhat superfluous. Of course, this difference may matter for proving certain technical corollaries. However, the main difference, in the author's opinion, is that the two approaches

often require two different proof techniques for establishing the limiting shape of the histogram of permuted test statistics. While the proof establishing almost sure convergence is typically more challenging, it is often different enough that if one is struggling to establish Hoeffding's criteria, it may be worthwhile to attempt establishing almost sure convergence instead. For a technical discussion and comparative analysis of convergence proofs, see Pouliot (2023).

Both approaches are in use. For example, in the main paper we used the Hoeffding criteria approach. So did Chung and Romano (2013). Meanwhile, Romano (1990), DiCiccio and Romano (2017) and D'Haultfœuille and Tuvaandorj (2024) establish almost sure convergence.

What we call the weak null is a sufficient condition for asymptotic validity of the randomization test. A randomization test is asymptotically valid if, under the null, the randomization distribution of its test statistic coincides, in arbitrarily large samples, with its resampling distribution. The randomized test then inherits, in such large samples, the validity of the resampling procedure.

The purpose of studentization is to further weaken the requirements of the weak null. This idea was first put forward by Neuhaus (1993) and was used in a similar context by D'Haultfœuille and Tuvaandorj (2024) and DiCiccio and Romano (2017); these also provide a more complete and historical set of references pertaining to studentization.

In standard settings, the resampling and permutational limit distributions of a test statistic are normal, but their variances may disagree. However, if there exists a variance estimate that converges to the correct, possibly different asymptotic variances in each asymptotic regime, then by studentizing the test statistic with that variance estimate, both limits of the studentized test statistic will be standard normal and asymptotic validity will obtain. As discussed above, this guarantees that the test is asymptotically valid, i.e., $E\left[\phi\left(\mathbf{W}\right)\right] \to \alpha$. This likewise guarantees the validity of other quantities of interest such as p-values. See Pouliot (2023) for further discussion.

6. DEFAULT SCHEDULE FOR THE NUMBER OF BLOCKS

We detail the tradeoffs involved in selecting the number n_{blocks} of blocks and provide a default schedule in terms of the number n of observations and the number p_2 of nuisance regressors. This schedule is implemented in the statistical package exactt.

The first-order tradeoff is between power and asymptotic robustness. On the one hand, more blocks mean more block permutations of $\mathbf{X}_2$ with respect to which $\mathbf{X}_1$ has to be orthogonalized in order to produce $\overline{\mathbf{X}}_1$, which in turn contains the explanatory variation on which the test relies for statistical power. In that sense, using more blocks has a cost in power. On the other hand, asymptotic robustness avails as the number of blocks increases; one can formally characterize the test as being more robust for a greater number of blocks.

The tradeoff with power is in fact a little more subtle because a larger number of blocks, and thus of permutations, makes optimization in preprocessing (see Section 7.1) more computationally expensive. Given a fixed computational budget or allowed preprocessing time, the power gains from preprocessing are traded off against robustness.

A greater number of nuisance regressors also means more variation in $\mathbf{X}_1$ has to be forgone in order to produce $\overline{\mathbf{X}}_1$. A good rule of thumb to not distort power, in light of Theorem 2 in the main text, is to scale n_{blocks} like $1/\sqrt{p_2}$, for a given sample size n.

While the theory suggests growth rates for the quantities of interest, levels still have to be established. These are informed by experience with real and synthetic data as well as by tradeoff with computational costs which are not captured by the theory.

The schedule of Table VI is a good reflection of practical experience. The test appears to perform well with as few as five blocks, while fewer than five blocks produces too granular a

TABLE VI

DEFAULT SCHEDULE OF n_{blocks} IN STATISTICAL PACKAGE EXACTT.

$p_2 \backslash n_{\text{blocks}}$	5	6	7	8	9	10	11
1	< 400	400	1000	1850	3000	4500	6000
2	< 565	565	1415	2615	4240	6370	8500
3	< 690	690	1730	3200	5200	7800	10000
4	< 800	800	2000	3700	6000	9000	12000
p_2	$< 100\sqrt{p_2}$	$400\sqrt{p_2}$	$1000\sqrt{p_2}$	$1850\sqrt{p_2}$	$3000\sqrt{p_2}$	$4500\sqrt{p_2}$	$6000\sqrt{p_2}$

randomization distribution, which may furthermore raise power concerns. With five blocks, we can optimize the block formation at moderate computational costs.

Meanwhile, large n applications are not uncommon. For instance, researchers sometimes favor randomization inference for its "placebo test" flavor even though they are analyzing a large sample. In such applications, in particular when the randomization hypothesis is implausible, we want to select a large number of blocks. We have found good performance using seven or eight blocks in large n simulations and applications. Nevertheless, in order to rely on asymptotic robustness and preserve against, for instance, heteroskedasticity, we take the "conservative" approach and further increase the number of blocks, n_{blocks}, as n increases. To be sure, the rates implied by the theory give but a loose guideline for picking the schedule. For tuning n_{blocks}, the key insight when moving from theory to practice is the sense in which we balance power against asymptotic robustness.

Finally, note that even if, past some value of n, we stop increasing the size of the random selection of permutations we use to implement the test in practice, there are still robustness benefits –and of course power costs– to increasing the number of blocks.

7. PREPROCESSING AND COMPUTATIONS

We expect the proposed test to be powerful if

$$\mathbf{X}_1^T \mathcal{Q}_{\mathbf{GX_2}} \left(\mathbf{Y} - \mathbf{X}_1 \beta_1^0 \right) = \mathbf{X}_1^T \mathcal{Q}_{\mathbf{GX_2}} \left(\mathbf{X}_1 \left(\beta_1 - \beta_1^0 \right) + \varepsilon \right) \tag{8}$$

is large when $\beta_1 - \beta_1^0$ is large. Intuitively, we have more power if the "projected out", or orthogonalized, vector $\overline{\mathbf{X}}_1 = \mathcal{Q}_{\mathbf{GX_2}} \mathbf{X}_1$ captures more of the variation of the tested covariate, i.e., when $\overline{\mathbf{X}}_1^T \mathbf{X}_1$ is large.

At first sight, this may seem like a daunting requirement for power. Indeed, for the candid choice of $\mathbf{G}$ the set of all permutations, we typically have $\overline{\mathbf{X}}_1 = \mathbf{0}_n$ and the test has no power whatsoever. However, and rather remarkably, for the non-overlapping block permutations for which the number of permutations $|\mathbf{G}|$ is factorial in the number of blocks, the rank of $\mathbf{GX_2}$, and thus the dimension of the space over which the $\mathbf{X}_1$ variation is annihilated in order to produce $\overline{\mathbf{X}}_1$, is only quadratic in the number of blocks. See Theorem 2 in the main paper for a precise statement. In other words, much power is retained even when we impose invariance in this specific way, for this specific group of linear actions.

7.1. *Preprocessing by optimizing ordering*

The order of the observations, or rather the choice of observations we select to form each block, impacts the value of (8) and in that sense matters for power. This suggests that we want,

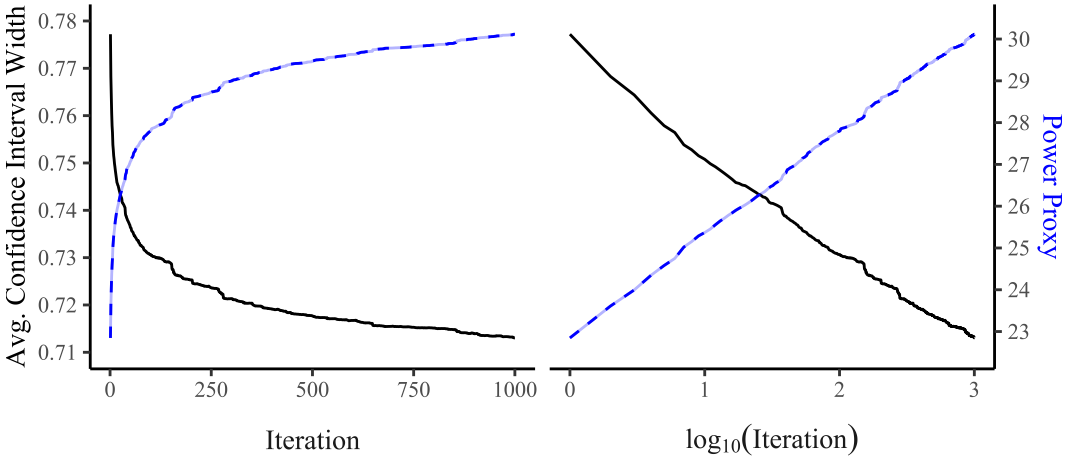


FIGURE 5.—Average criteria evaluation and confidence interval width versus number of iterations and $\log_{10}$ of the number of iterations.

in preprocessing, to order the observations such that our proxy for power is maximized, and that our statistical package always outputs the same interval for the same dataset, regardless of the order in which the observations are arranged originally.

Let Π be the set of all permutations from and to $\{1, \ldots, n\}$. We solve for the optimal reordering π^* of the data. Specifically, we solve

$$\pi^* = \arg\max_{\pi \in \Pi} \left\{ \mathbf{X}_{\pi,1}^T \mathcal{Q}_{\mathbf{G}\mathbf{X}_{\pi,2}} \mathbf{X}_{\pi,1} - \text{mean}_{g \in \mathbf{G}_n} \left(\mathbf{X}_{\pi,1}^T \mathcal{Q}_{\mathbf{G}\mathbf{X}_{\pi,2}} \mathbf{X}_{g\pi,1} \right) \right\}. \tag{9}$$

This is akin to, and very much in the same spirit as Section 2.4, "Pre-ordering rows of design matrix" from Lei and Bickel (2021). It is likewise the analog of the permutation set construction of Wen et al. (2022). Note that the second summand is there to protect against shifting the randomization distribution too much to the right, thus forgoing the power gains from shifting the realized test statistic to the right. This addition seems to make an appreciable difference in practice with smaller samples.

As in Lei and Bickel (2021), we solve this program using off-the-shelf software implementing a genetic algorithm. Specifically, we use the R CRAN package GA.

However, since both Lei and Bickel (2021) and Wen et al. (2022) only proceed under the randomization hypothesis, they are not concerned with the validity of the test after preprocessing in the presence of heteroskedasticity. The question is more nuanced in our case since, under heteroskadasticity, the preprocessing routine (9) could be adapting to variation in the outcome variable $\mathbf{Y}$.[4]

Fortunately, the preprocessed test statistic has value under the null $n^{-1/2} \mathbf{X}_{\pi,1}^T \mathcal{Q}_{\mathbf{G}\mathbf{X}_{\pi,2}} \varepsilon_\pi = n^{-1/2} \mathbf{X}_{\pi,1}^T \varepsilon_\pi + o_p(1)$, if $n_{\text{blocks}}^4 / n \to 0$. Consequently, the blocking does not inform the asymptotic null distribution of the test statistic.

7.2. *Computations*

A standard implementation of test inversion would first fix a grid of candidate null values for β_1 and evaluate the hypothesis test for each value.

[4]I thank Kei Hirano for this observation.

It is in fact possible to extract the discrete set of all values of β_1^0 at which, if we were continuously increasing β_1^0, the p-value associated with the null hypothesis $H_0 : \beta_1 = \beta_1^0$ would change. Obviously, these are all the values at which we need to evaluate the p-value in order to compute the complete map from $\beta_1^0 \in \mathbb{R}$ to its associated p-values.

Furthermore, one need not implement the entire test procedure at each such candidate β_1^0, but can in fact proceed sequentially with simple updates. Such a test inversion procedure is laid out and further generalized in Xu (2025), which also provides geometric intuition.

For simplicity of exposition, write

$$T_g\left(\beta_1\right) = a_g + b_g \beta_1^0,$$

where $a_g = \overline{\mathbf{X}}_1^T \left(g\mathbf{Y}\right)/\hat{\sigma}_g$ and $b_g = -\overline{\mathbf{X}}_1^T \left(g\mathbf{X}_1\right)/\hat{\sigma}_g$ for each $g \in \mathbf{G}_n$. We state the procedure for $\tilde{\mathbf{G}}_n \subset \mathbf{G}_n$, a random subset of $\mathbf{G}_n$. Let $\tilde{M}$ denote the cardinality of $\tilde{\mathbf{G}}_n$.

Algorithm 1 Sequential Inversion

Preprocessing:
Compute $\left\{a_g, b_g\right\}_{g \in \tilde{\mathbf{G}}_n}$
Compute and sort $\beta_1^{(1)} \leq \beta_1^{(2)} \leq \cdots \leq \beta_1^{(\tilde{M})}$, where $\beta_1^{(j)} = \frac{a_{g(j)} - a_{\mathrm{id}}}{b_{\mathrm{id}} - b_{g(j)}}$
Compute $p^{(1)} = \frac{1}{\tilde{M}} \sum_{j=1}^{\tilde{M}} \mathbf{1}\left\{a_{\mathrm{id}} + b_{\mathrm{id}}\beta_1^0 \leq a_{g^{(1)}} + b_{g^{(1)}}\beta_1^0\right\}$
for j in $2, \ldots, \tilde{M}$ **do**
    if $b_{g^{(j)}} < b_{\mathrm{id}}$ **then**
        $p^j = p^{j-1} - \frac{1}{\tilde{M}}$
    else
        $p^j = p^{j-1} + \frac{1}{\tilde{M}}$
    end if
end for

One may for instance plot $\left(p^{(1)}, \ldots p^{(\tilde{M})}\right)$ against $\beta_1^{(1)} \leq \beta_1^{(2)} \leq \cdots \leq \beta_1^{(\tilde{M})}$. In particular, if the cardinality of $\mathbf{G}_n$ allows one to run the algorithm with $\tilde{\mathbf{G}}_n = \mathbf{G}_n$, the interpolated plot of the computed p-values against the null coefficients candidate will represent the exhaustive map from $\beta_1^0 \in \mathbb{R}$ to the p-value of $H_0 : \beta_1 = \beta_1^0$.

Algorithm 1 details the inversion procedure assuming, for ease of exposition, that the "slopes" b_g are never equal to b_{id}. For more details, see Xu (2025).

8. ADDITIONAL INTUITION FOR THE EXACT VALIDITY VIA COUNTEREXAMPLES

Preliminary intuition may be teased out from the univariate case $\mathbf{Y} = \mathbf{1}_n + \mathbf{X}_1\beta_1 + \eta$, where again we want to test $H_0 : \beta_1 = \beta_1^0$. A natural test statistic is $T_g^{\mathrm{simple}}\left(\beta_1^0\right) = \mathbf{X}_1^T\left(g\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)\right)$ for g in the set of all permutations on $\{1, \ldots, n\}$. The corresponding permutation test is valid under the strong null stating that $\left(\mathbf{X}_1, \varepsilon\right) \sim \left(\mathbf{X}_1, g\varepsilon\right)$, for all permutations g. The test has nontrivial power because the signal component $\mathbf{X}_1^T\left(\mathbf{X}_1\left(\beta_1 - \beta_1^0\right)\right)$ is larger for larger values of $|\beta_1 - \beta_1^0|$, while $\mathbf{X}_1^T\left(g\mathbf{X}_1\left(\beta_1 - \beta_1^0\right)\right)$ remains weakly smaller given $\beta_1 - \beta_1^0$ for any $g \neq \mathrm{id}$, because $\mathbf{X}_1^T g \mathbf{X}_1$ is weakly smaller than $\mathbf{X}_1^T\mathbf{X}_1$.

We want to produce such an exact test in the multivariate case $\mathbf{Y} = \mathbf{1}_n + \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \varepsilon$. Obviously, $T_g^{\mathrm{simple}}\left(\beta_1^0\right) = \mathbf{X}_1^T\left(g\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)\right) = \mathbf{X}_1^T\left(g\mathbf{X}_2\beta_2\right) + \mathbf{X}_1^T\varepsilon$ is not invariant to β_2,

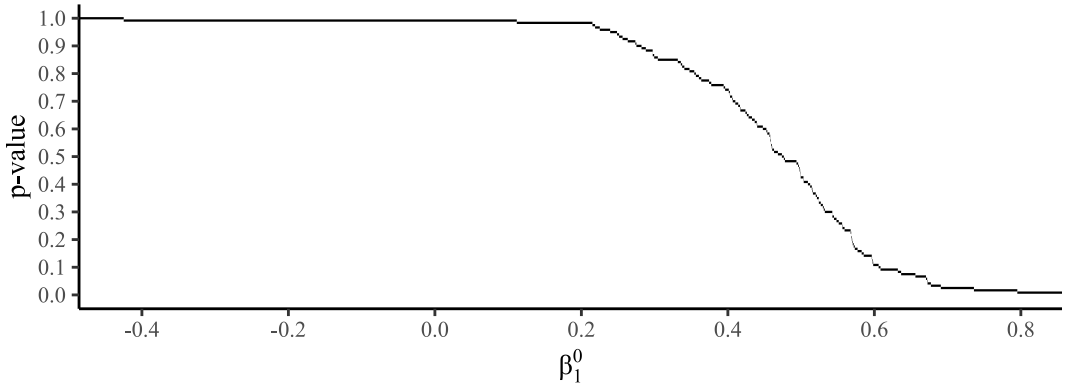


FIGURE 6.—p-value vs β_1^0, from Xu (2025).

and as such is not a candidate for a valid test unless X_{i1} and X_{i2} are independent. Concretely, if X_1 and X_2 were, say, positive correlated, $\mathbf{X}_1^T\mathbf{X}_2\beta_2$ would tend to be bigger than $\mathbf{X}_1^T(g\mathbf{X}_2\beta_2)$, and the procedure would have excess probability of Type I error.

One may consider the candidate statistic $T_g^{(a)}(\beta_1^0) := \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}(g(\mathbf{Y}-\mathbf{X}_1\beta_1^0))$ since

$$T_{\mathrm{id}}^{(a)}\left(\beta_1^0\right) := \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right) = \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}\left(\mathbf{X}_2\beta_2+\varepsilon\right) \overset{H_0}{=} \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}\varepsilon,$$

which is invariant with respect to β_2. However, typically $\mathcal{Q}_{\mathbf{X}_2}g\mathbf{X}_2\beta_2 \neq 0$ and $T_{\mathrm{g}}^{(a)}(\beta_1^0)$ is not invariant for $g \neq \mathrm{id}$, thus the testing procedure is not valid; $T_{\mathrm{id}}^{(a)}(\beta_1^0)$ would typically not even have the same marginal resampling distribution as $T_g^{(a)}(\beta_1^0)$ for $g \neq \mathrm{id}$.

Conversely, the likewise *a priori* tempting candidate $T_g^{(b)}(\beta_1^0) := (g\mathbf{X}_1)^T\mathcal{Q}_{\mathbf{X}_2}(\mathbf{Y}-\mathbf{X}_1\beta_1^0)$ is invariant with respect to β_2 for all group actions g. However, considering the form of the test statistic under the null,

$$T_g^{(b)}\left(\beta_1^0\right) \overset{H_0}{=} (g\mathbf{X}_1)^T\mathcal{Q}_{\mathbf{X}_2}\varepsilon,$$

it is easy to see that this is a function of $(g\mathbf{X}_1, \mathbf{X}_2, \varepsilon)$ and that the implicit strong null required for the exact validity of the test is $(\mathbf{X}_1, \mathbf{X}_2, \varepsilon) \sim (g\mathbf{X}_1, \mathbf{X}_2, \varepsilon)$, for all g in a specified group. For, say, a group of permutations, this essentially requires that the tested covariate X_{i1} be independent of the remaining covariates X_{i2}, which is not the case of practical interest for multivariate linear regression with observational data. In fact, this case can be tackled with the aforementioned test for univariate regression by simply considering $\eta = \mathbf{X}_2\beta_2 + \varepsilon$ as the error vector.[5]

These counterexamples are suggestive in two ways. One interpretation key is that the test statistic must be invariant with respect to $g\mathbf{X}_2\beta_2$ for any g in a selected group of linear actions $\mathbf{G}$. A choice of orthogonal projection matrix accomplishing this is $\mathcal{Q}_{\mathbf{GX_2}}$ where $\mathbf{GX_2} = (g\mathbf{X}_2)_{g\in\mathbf{G}}$, the matrix whose columns are $g\mathbf{X}_2$ for all $g \in \mathbf{G}$. Trivially, $\mathcal{Q}_{\mathbf{GX_2}}g\mathbf{X}_2 = 0$ for all $g \in \mathbf{G}$, and under the null,

$$T_g\left(\beta_1^0\right) := \mathbf{X}_1^T\mathcal{Q}_{\mathbf{GX_2}}\left(g\left(\mathbf{Y}-\mathbf{X}_1\beta_1^0\right)\right) = \mathbf{X}_1^T\mathcal{Q}_{\mathbf{GX_2}}\left(g\left(\mathbf{X}_2\beta_2+\varepsilon\right)\right) = \mathbf{X}_1^T\mathcal{Q}_{\mathbf{GX_2}}\left(g\varepsilon\right).$$

[5]A similar observation holds for $T_g^{(c)}(\beta_1^0) := \mathbf{X}_1^T\mathcal{Q}_{g\mathbf{X}_2}(g(\mathbf{Y}-\mathbf{X}_1\beta_1^0)) \overset{H_0}{=} \mathbf{X}_1^T\mathcal{Q}_{g\mathbf{X}_2}(g\varepsilon)$.

The test using the statistic $T_g\left(\beta_1^0\right)$ is valid under the strong null presented in Assumption 2 of the main text for a general group of linear actions $\mathbf{G}$. Indeed, we emphasize that the argument holds for any group $\mathbf{G}$ of transformations. See Dutz and Zhang (2025) for a discussion.

Remark that $T_g\left(\beta_1^0\right)$ is precisely the numerator of the proposed test statistic (4) in the main text. As detailed in Section 3 in the main text, the denominator is introduced for purposes of asymptotic robustness and its addition does not compromise exact validity.

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